

Beta 0.7 technical reference

This is the authoritative technical description of the released diagnostic. It combines the former address-map, 16 KiB ROM, firmware-checksum, and Beta 0.7 specification notes.

Address map

CPU address	Hardware or content	Notes
>0000->1FFF	Console CPU ROM	Read twice for stability and identity
>6000->7FFF	Diagnostic extension	Cartridge header, menu, and interactive tests
>8300->83FF	Common scratchpad	Saved, tested, and restored
>8400	Sound write port	Muted at startup; driven by the PSG test
>8800 , >8802	VDP read/status	Direct hardware access
>8C00 , >8C02	VDP write/control	Direct hardware access
>9800 , >9C02	GROM read/address	Stable-read and identity checks only
>C000->DFFF	Tester U7 SRAM	Workspace, results, and relocated demo code
>E000->FFFF	Fixed diagnostic core	Automatic tests and LOAD recovery
>E002 on writes	Six-bit LED latch	Direction decode separates writes from ROM reads
>FFFC->FFFF	LOAD vector	Workspace >C000 , entry >FF00

The ordinary TI 32K expansion also drives >C000->FFFF and must not be present while the tester owns U7 SRAM and U3 ROM. The raster controller runs from the tester's own U7 SRAM at >D000->D537 ; it does not require optional RAM at >A000 .

W27C512 layout

The tested 16 KiB modification is direct mapping, not bank switching. The 65,536-byte programmer file is sparse:

File offsets / CPU addresses	Content
>6000->7FFF	8 KiB interactive extension
>E000->FFFF	8 KiB automatic core
all other offsets	>FF

The fixed core validates the extension signature, version, entry range, ABI, and zero word sum before calling it. A missing or corrupt extension cannot prevent the automatic tests, LED report, LOAD path, or fixed-core result page.

The extension begins with a standard TI-99/4A cartridge header. Its menu entry, `HEXBUS DIAG 1.0` , executes `BLWP @>FFFC` , the same vector used by the physical LOAD switch. A TI-99/4 does not use that cartridge menu and starts through LOAD.

Automatic test coverage

Target	Coverage
TMS9900	Representative arithmetic, logic, shift, compare, and branch paths
Tester SRAM	>C100->DFFF , excluding active workspace/result storage
Scratchpad	>8300->83FF , save/test/restore
VRAM	All >0000->3FFF bytes with 00 , FF , AA , and 55
Console CPU ROM	>0000->1FFF , forward and reverse reads
Diagnostic core	Complete >E000->FFFF zero word sum
System GROMs	GROM 0, 1, and 2, two 6 KiB reads each
LED latch	Six-output walk followed by settled PASS/FAIL groups

The VRAM test is destructive. A failure or power interruption cannot promise that scratchpad state will be restored.

Firmware identity and read health

Read health and firmware identity are deliberately separate. Two reads that do not agree are a diagnostic failure. A stable custom, replacement, or mixed ROM set is reported as information and is not colored or labeled as failed.

Profile	CPU ROM word sum	GROM 0 byte sum	GROM 1	GROM 2
TI-99/4 (1979)	>F1BB	>B16C	>1B7B	>9EC3
TI-99/4A (1981)	>7D8C	>5574	>1B99	>9ADC
TI-99/4QI V2.2 (1983)	>7D8C	>31D7	>1B99	>9ADC

The CPU value is the modulo-65536 sum of 4096 big-endian words. Each GROM value is the modulo-65536 sum of 6144 bytes. These compact runtime identifiers are not catalog-file CRC or SHA hashes.

Keyboard and joystick ownership

The diagnostic does not call the console ROM keyboard vector at >000E and does not call a GROM routine. It scans the TMS9901 directly, using separate translation tables for the five-column TI-99/4 keyboard and six-column TI-99/4A keyboard. Unknown or GROMless recovery systems default to the /4A table so menu input remains available.

The TI-99/4 uses joystick selector columns 5 and 6. The TI-99/4A and QI use columns 6 and 7. The page reports modifier state separately from translated characters, recognizes two-key chords plus Alpha Lock, and never executes QUIT or other console actions while the diagnostic is active.

Video and sprite contract

The TI-99/4 skips the Graphics-II bitmap stage because its original TMS9918 does not support that mode. Text, character, Multicolor, sprite, and raster tests remain available. Operator-facing screens say VDP ; they do not infer a specific VDP model from display behavior.

The sprite exercise writes five same-scanline entries to the sprite attribute table at `>0300`, using pattern table `>1800`. Sprites 0 and 1 overlap for the collision condition; sprites 2 and 3 are separate; sprite 4 exercises the four-per-line limit. The expected original-family display is three visible groups, with collision and fifth-sprite flags set. Replacement VDPs may differ, so the result page reports flags as observations rather than pass/fail claims.

Every return from a video exercise blanks the display, restores all eight UI registers, rebuilds the font, colors, frame, and sprite terminator, and only then enables display. This ownership boundary prevents demo sprites or table selections from corrupting later pages.

Beta 0.7 qualification

The exact public programmer image passed the static acceptance checks and the 59-case internal emulator regression suite. It was then exercised on stock TI-99/4A, TI-99/4QI V2.2, GROMless TI-99/4A LOAD recovery, and a TI-99/4 fitted with a Pico9918. This is a public beta because the sample of original and replacement VDP hardware remains limited.